

| Newton on KKT                                                                                                                                                                                                             | KKT Analytical                                                                                                                                                                                                           |
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| $\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = 4 - x_1 - x_2 \leq 0$                                                                                                                                                   | $\min f(x) = x_1^2 + 5x_2^2, \text{ s.t. } g(x) = x_1 + x_2 - 4 \leq 0$                                                                                                                                                  |
|                                                                                                                                                                                                                           | <b>Slack:</b> $g(x) - s^2 = 0$ , so $l(x, \lambda, s) = f(x) - \lambda(g(x) - s^2)$                                                                                                                                      |
|                                                                                                                                                                                                                           | $\frac{\partial l}{\partial s} = 2\lambda s = 0 \Rightarrow \lambda = 0$ (inactive) or $s = 0$ (active)                                                                                                                  |
|                                                                                                                                                                                                                           | <b>Test:</b> $X_{\text{unc}} = (0, 0)$ ; $g(X_{\text{unc}}) = -4 < 0 \quad \therefore$ infeasible $\Rightarrow \lambda \neq 0$<br>$\xrightarrow{2\lambda s=0} s = 0$ , constraint <b>active</b>                          |
| $l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (same $l$ , implicit in $\nabla_x l$ )                                                                                                                          | $l(x, \lambda) = x_1^2 + 5x_2^2 - \lambda(x_1 + x_2 - 4)$ (set $s = 0$ )                                                                                                                                                 |
| Assert $r(x, \lambda) = 0$ and $J$ ; note $J = \begin{bmatrix} H_l & -\nabla_x g \\ \nabla_x g^\top & 0 \end{bmatrix}$ :                                                                                                  | Same $r$ and $J$ arrived at from $\nabla_x l = 0$ and $g = 0$ :                                                                                                                                                          |
| $r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$  | $r = \begin{bmatrix} \nabla_x l \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} \quad J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$ |
| <b>Init:</b> $z_0 = (0, 0, 0)^\top$ ; evaluate $r$ at $z_0$ :                                                                                                                                                             | (no iterate — constraint known active from test above)                                                                                                                                                                   |
| $r(z_0) = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix}_{z_0=(0,0,0)} = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$ |                                                                                                                                                                                                                          |
| <b>Newton step:</b> $J \Delta z = -r(z_0)$ , i.e. $-r(z_0) = (0, 0, 4)^\top$ :                                                                                                                                            | $\nabla_x l = H_f x - \lambda \nabla_x g = 0 \Rightarrow x^*(\lambda) = H_f^{-1} \lambda \nabla_x g$                                                                                                                     |
| $J \Delta z = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$                  | $\begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \end{bmatrix} = 0 \Rightarrow x^*(\lambda) = \begin{bmatrix} \lambda/2 \\ \lambda/10 \end{bmatrix}$                                                                   |
| Row 1: $2\Delta x_1 - \Delta \lambda = 0 \Rightarrow \Delta x_1 = \frac{\Delta \lambda}{2}$<br>Row 2: $10\Delta x_2 - \Delta \lambda = 0 \Rightarrow \Delta x_2 = \frac{\Delta \lambda}{10}$                              |                                                                                                                                                                                                                          |
| Row 3: $\frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6\Delta \lambda = 40 \Rightarrow \Delta \lambda = \frac{20}{3}$                                                                              | Sub $x^*(\lambda)$ into $g = 0$ : $\frac{\lambda}{2} + \frac{\lambda}{10} = 4 \Rightarrow 10\lambda + 2\lambda = 80 \Rightarrow \lambda = \frac{20}{3}$                                                                  |
| $\Delta x_1 = \frac{10}{3}, \Delta x_2 = \frac{2}{3}; z_1 = z_0 + \Delta z = (\frac{10}{3}, \frac{2}{3}, \frac{20}{3})$                                                                                                   | $x^* = \left(\frac{10}{3}, \frac{2}{3}\right)^\top$                                                                                                                                                                      |
| $f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$                                                                                                        | $f(x^*) = \left(\frac{10}{3}\right)^2 + 5\left(\frac{2}{3}\right)^2 = \frac{100}{9} + \frac{20}{9} = \frac{40}{3}$                                                                                                       |

**SQP connection**

**KKT** — necessary conditions for a constrained optimum; gives the system  $r(x, \lambda) = 0$

**Lagrangian** — function whose stationarity encodes the KKT conditions; introduces  $\lambda$

**Slack variables** — convert inequalities to equalities; lead to complementary slackness  $\lambda s = 0$

**SQP** — solves  $r = 0$  using Newton’s method; each step inverts the Jacobian of  $r$ :

$$r(x, \lambda) = \begin{bmatrix} \nabla_x \mathcal{L} \\ g(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - \lambda \\ 10x_2 - \lambda \\ x_1 + x_2 - 4 \end{bmatrix} = 0 \quad J = \frac{\partial r}{\partial (x_1, x_2, \lambda)} = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

**BFGS** — quasi-Newton approximation of the Hessian of  $\mathcal{L}$  when exact computation is too costly

**Newton’s method applied to KKT** (same example:  $\min x_1^2 + 5x_2^2$  s.t.  $x_1 + x_2 = 4$ , starting at  $z_0 = (0, 0, 0)$ )

**Step 1 — residual at  $z_0$ :**

$$r(z_0) = \begin{bmatrix} 2(0) - 0 \\ 10(0) - 0 \\ 0 + 0 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4 \end{bmatrix}$$

**Step 3 — solve  $J \Delta z = -r(z_0)$ :**

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

**Step 2 — Jacobian** (constant here since  $r$  is linear):

$$J = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 10 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

Row 1:  $2\Delta x_1 - \Delta \lambda = 0 \Rightarrow \Delta x_1 = \frac{\Delta \lambda}{2}$   
Row 2:  $10\Delta x_2 - \Delta \lambda = 0 \Rightarrow \Delta x_2 = \frac{\Delta \lambda}{10}$   
Row 3:  $\Delta x_1 + \Delta x_2 = 4 \Rightarrow \frac{\Delta \lambda}{2} + \frac{\Delta \lambda}{10} = 4 \Rightarrow 6\Delta \lambda = 40 \Rightarrow \Delta \lambda = \frac{20}{3}$

$$\Delta x_1 = \frac{10}{3} \quad \Delta x_2 = \frac{2}{3}$$

**Step 4 — update:**  $z_1 = z_0 + \Delta z$

$$z_1 = \left(\frac{10}{3}, \frac{2}{3}, \frac{20}{3}\right) = x^* \quad \checkmark \quad (\text{one step: KKT system is linear})$$

**Newton — unconstrained**     $p_0 = -H_f(X_0)^{-1}\nabla f(X_0)$   
 $f(x_1, x_2) = x_1^2 + 5x_2^2, \ X_0 = (3, 1)^\top$

$$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 10x_2 \end{bmatrix} \Rightarrow \nabla f(X_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 10 \end{bmatrix} \Rightarrow H_f^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix}$$

$$p_0 = - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

**Line search:**  $\hat{X}_1 = \begin{bmatrix} 3 - 3\alpha \\ 1 - \alpha \end{bmatrix}$

$$\phi(\alpha) = (3 - 3\alpha)^2 + 5(1 - \alpha)^2 = 14 - 28\alpha + 14\alpha^2$$

$$\frac{d\phi}{d\alpha} = -28 + 28\alpha = 0 \Rightarrow \alpha_0 = 1$$

**Update:**

$$X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 1 \cdot \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$f(X_1) = 0$  — one step, exact (quadratic  $f$ ,  $\alpha = 1$  always)

**BFGS — quasi-Newton**     $p_0 = -B_0^{-1}\nabla f(X_0), \ B_0 = I$   
 $f(x_1, x_2) = x_1^2 + 5x_2^2, \ X_0 = (3, 1)^\top$

$$\nabla f(X_0) = \begin{bmatrix} 6 \\ 10 \end{bmatrix} \Rightarrow p_0 = -I \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -6 \\ -10 \end{bmatrix}$$

**Line search:**  $\hat{X}_1 = \begin{bmatrix} 3 - 6\alpha \\ 1 - 10\alpha \end{bmatrix}$

$$\phi(\alpha) = (3 - 6\alpha)^2 + 5(1 - 10\alpha)^2 = 14 - 136\alpha + 536\alpha^2$$

$$\frac{d\phi}{d\alpha} = -136 + 1072\alpha = 0 \Rightarrow \alpha_0 = \frac{136}{1072} \approx 0.1269$$

**Update:**

$$X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 0.1269 \begin{bmatrix} -6 \\ -10 \end{bmatrix} = \begin{bmatrix} 2.239 \\ -0.269 \end{bmatrix}$$

**BFGS vectors** (for  $B_1$  update):

$$s_0 = X_1 - X_0 = \begin{bmatrix} -0.7614 \\ -1.269 \end{bmatrix}$$

$$y_0 = \nabla f(X_1) - \nabla f(X_0) = \begin{bmatrix} 4.478 \\ -2.69 \end{bmatrix} - \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1.522 \\ -12.689 \end{bmatrix}$$

$$s_0, y_0 \text{ feed the rank-2 update: } B_1 = B_0 - \frac{B_0 s_0 s_0^\top B_0}{s_0^\top B_0 s_0} + \frac{y_0 y_0^\top}{y_0^\top s_0}$$